1 A note on notations

1.1 Material derivative

The following equations are all equivalent, which can be shown algebraically.

$$\begin{aligned}\frac{d}{dt} &= \frac{\partial}{\partial t} + u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} + w\frac{\partial}{\partial z} \\ &= \frac{\partial}{\partial t} + \mathbf{u}\cdot\nabla \\ &= \frac{\partial}{\partial t} + \mathbf{u}\cdot\mathrm{div}()\end{aligned}$$

1.2 Tensors

$$\tau_{ij}\frac{\partial u_i}{\partial x_j} = \boldsymbol{\tau} : \nabla\mathbf{u}$$

where

$$\nabla\mathbf{u} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix}$$

The : is called the double contraction operator or the matrix inner product. It contracts two tensors into a scalar. Since τ is the symmetric stress tensor,